# CH 3 — Diagonalizability

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Warm Up Question

Let $R : \mathbb{R}^2 \to \mathbb{R}^2$ be reflection in the line $y = 3x$. Determine $R(\vec{v})$

Reflection is a linear map. $R$ can be represented by a $2 \times 2$ matrix.

Choose a basis $B = \{e_1, e_2\} \in \mathbb{R}^2$. Build ${}_B[A]_B$

$A = [R]_B = \begin{bmatrix} [R(e_1)]_B & [R(e_2)]_B \end{bmatrix}$. But we don't know $R(e_1), R(e_2)$.

We choose instead the vector $R(v_1) = v_1$, that is on the line $y = 3x$, say $v_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

Then choose another vector $R(v_2) = -v_2$, that is perpendicular to the line $y = 3x$, say $v_2 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$

$D = \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \end{bmatrix} \right\}$

Now $C = {}_D[R]_D = \begin{bmatrix} [R(v_1)]_D & [R(v_2)]_D \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

**Lesson**:

1. Sometimes there may be a "nice" basis for $L : V \to V$ that makes computations easier
2. Dream scenario: want the basis $D$ so that we have $[L]_D = \begin{bmatrix} x & \dots & 0 \\ \dots & \dots & \dots \\ 0 & \dots & x \end{bmatrix}$

> 📘 **Definition — Operator**
>
> Let $T : V \to W$ is called a **linear operator** if $\dim V = \dim W$

> 📘 **Definition — Diagonalizability**
>
> Compute the matrix of $L : V \to V$ and obtain the basis $D$ so that we have $[L]_D = \begin{bmatrix} x & \dots & 0 \\ \dots & \dots & \dots \\ 0 & \dots & x \end{bmatrix}$
>
> If there is such a basis $D$, we say $L$ is **diagonalizable** and we say $D$ is a **diagonalizable basis**

Example:

Reflection map $R : \mathbb{R}^2 \to \mathbb{R}^2$ across $y = 3x$ is diagonalizable and $D = \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \end{bmatrix} \right\}$ is a diagonalizable basis.

Suppose we want to calcualte $R(\vec{v})$.

$[R(\vec{v})]_D = [R]_D[\vec{v}]_D$

But we want $R(\vec{v}) = [R(\vec{v})]_S, S = \{e_1, e_2\}$

Let ${}_SI_D$ be change for $D$ to $S$ coordinates. Then ${}_SI_D[\vec{x}]_D = [\vec{x}]_S$

We obtain

$$
\begin{aligned}
{}_S I_D[R(\vec{v})]_D &= {}_S I_D[R]_D[\vec{v}]_D \\
[R(\vec{v})]_S &= {}_S I_D[R]_D[\vec{v}]_D \\
[R(\vec{v})]_S &= {}_S I_D[R]_D({}_S I_D)^{-1}{}_S I_D[\vec{v}]_D \\
[R(\vec{v})]_S &= {}_S I_D[R]_D({}_S I_D)^{-1}[\vec{v}]_S
\end{aligned}
$$

So

$$
{}_S[R]_S = {}_S I_D{}_D[R]_D{}_D I_S
$$

${}_S I_D = \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}$

$\therefore [R]_S = \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\left(\begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}\right)^{-1} = \frac{1}{10}\begin{bmatrix} -8 & 6 \\ 6 & 8 \end{bmatrix}$

📘 **Definition — Similarity**

Let $A, B \in M_{n\times n}(\mathbb{F})$, we say $A$ is **similar** to $B$ if $\exists$ invertible matrix $P$ s.t. $A = PBP^{-1}$

If $A \sim B$

1. rank $A$ = rank $B$
2. nullity $A$ = nullity $B$
3. det $A$ = det $B$
4. tr $A$ = tr $B$

Example:

If $B, C$ are bases for $V$ and if $L : V \to V$ is linear, then ${}_B[L]_B \sim {}_C[L]_C$

since ${}_B[L]_B = {}_B I_C{}_C[L]_C({}_B I_C)^{-1}, P = {}_B I_C$

QUESTION: Is $[L]_B$ for any choice of $B$, similar to a diagonal matrix?

Warm Up

Suppose $D = v_1, ..., v_n$ is a diagonalizable basis for $L$. That is suppose $[L]_D = \begin{bmatrix} \lambda_1 & ... & 0 \\ ... & ... & ... \\ 0 & ... & \lambda_n \end{bmatrix}$

$\iff \left[[L(v_1)]_D ... [L(v_n)]_D\right] = \begin{bmatrix} \lambda_1 & 0 & ... & 0 \\ 0 & \lambda_2 & ... & 0 \\ ... & ... & ... & ... \\ 0 & ... & 0 & \lambda_n \end{bmatrix}$

Equating cols, $[L(v_1)] = \begin{bmatrix} \lambda_1 \\ 0 \\ ... \\ 0 \end{bmatrix} \iff L(v_1) = \lambda_1 v_1 + 0v_2 + ... + 0_n, \therefore L(v_1) = \lambda_1 v_1$

Similarly, $L(v_1) = \lambda_i v_i, i = 1, 2, ..., n$

So: $D = \{v_1, ..., v_n\}$ is a diagonalizable basis for $L$ if and only $L(v_1) = \lambda_i v_i, \forall i$ if and only $v_i$ is an eigenvector for $L$ with eigenvalue $\lambda$

📘 **Definition — Eigenbasis**

An **eigenbasis** for $L$ is a basis $\{v_1, ..., v_n\}$ where the $v_i$ are Eigenvectors.

(In this case $[L]_D = \begin{bmatrix} \lambda_1 & ... & 0 \\ ... & ... & ... \\ 0 & ... & \lambda_n \end{bmatrix}$ and the i-th diagonal entry $\lambda_i$ is the eigenvalue of $v_i$)

Examples:

1. $R : \mathbb{R}^2 \to \mathbb{R}^2$ reflection across $y = 3x$

   We found $v_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ is on the line and $R(v_1) = v_1$, so $v_1$ is an eigenvector with $\lambda_1 = 1$

   Also, $v_2 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$ is perpendicular to the line and $R(v_2) = -v_2$ is an eigenvector with $\lambda_2 = -1$

   Therefore, $D = \{v_1, v_2\}$ is an eigenbasis. We computed $[L]_D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ as expected.

2. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be rotation by $\theta \neq \pi, 2\pi, 0 < \theta < 2\pi$

   Then $T$ has no eigenvectors so $T$ has no eigenbasis and $T$ is not diagonalizable

3. Let $D : P_2(\mathbb{R}) \to P_2(\mathbb{R}), D(p(x)) = p'(x)$

   Does $D$ have any Eigenvectors

   Yes, if $p(x) = c, D(p(x)) = 0 = 0 \cdot p(x)$ is an eigenvector with $\lambda = 0$

   If $\deg p > 1$, then $D(p) \neq \lambda p, \lambda \neq 0$.

   The only possible eigenvalue is $\lambda = 0$ when eigenvector are constant.

   $\therefore D$ is not diagonalizable.

   Note, if $D$ : functions $\to$ functions, $D\left(e^{\lambda x}\right) = \lambda e^{\lambda x}$

4. Let $D : P_2(\mathbb{R}) \to P_2(\mathbb{R}), D(p(x)) = p'(x)$

   We work with matrix, $S = \{1, x, x^2\}$

   $[D]_S = \left[ [D(1)]_S \ [D(x)]_S \ [D(x^2)]_S \right] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$

   What is the relationship between eigenvectors and eigenvalues of $D$ and those of $A = [D]_S$?

   $D(p) = \lambda p \implies [D(p)]_S = [\lambda p]_S \iff [D]_S[p]_S = \lambda[p]_S \iff A[p]_S = \lambda[p]_S$

   $\therefore D(p) = \lambda p \iff A[p]_S = \lambda[p]_S$

📐 **Theorem —**

Let $L : V \to V$ be an operator. Let $B$ be a basis for $V$, $A = [L]_B$

Then $v \in V$ is an eigenvector for $L$ with eigenvalue $\lambda \iff [v]_B$ is an eigenvector for $A$ with eigenvalue $\lambda$

💡 **Tip** — Linear Operators

Whenever struggling to solve a question with linear operators, translate into matrix language and attempt to solve it.

Example:

$D : P_2(\mathbb{R}) \to P_2(\mathbb{R}), A = [D]_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$.

Notice this is a triangular matrix, the eigenvalue are the diagonal entries.

$\lambda = 0$ is the only eigenvalue for $A$

Therefore $\lambda = 0$ is the only eigenvalue for $D$

$Av = \lambda v \iff Av - \lambda v = 0 \iff (A - \lambda I)v = 0 \iff v \in \text{Null}(A = \lambda I)$

Eigenvectors $\begin{bmatrix} c \\ 0 \\ 0 \end{bmatrix}$